

[MS-RDPEWA]:

Remote Desktop Protocol: WebAuthn Virtual Channel Protocol

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1 Introduction

The Remote Desktop Protocol (RDP): WebAuthn Virtual Channel Protocol provides a way for a user to do WebAuthn operations over the RDP protocol. It enables a server to send webauthn request to a client, the client can then use this request to talk to authenticators (platform as well as cross-platform) and reply with the response.

Sections 1.5, 1.8, 1.9, 2, and 3 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

globally unique identifier (GUID): A term used interchangeably with universally unique identifier (UUID) in Microsoft protocol technical documents (TDs). Interchanging the usage of these terms does not imply or require a specific algorithm or mechanism to generate the value. Specifically, the use of this term does not imply or require that the algorithms described in [\[RFC4122\]](#) or [\[C706\]](#) have to be used for generating the GUID. See also universally unique identifier (UUID).

Remote Desktop Protocol (RDP): A multi-channel protocol that allows a user to connect to a computer running Microsoft Terminal Services (TS). RDP enables the exchange of client and server settings and also enables negotiation of common settings to use for the duration of the connection, so that input, graphics, and other data can be exchanged and processed between client and server.

Transmission Control Protocol (TCP): A protocol used with the Internet Protocol (IP) to send data in the form of message units between computers over the Internet. TCP handles keeping track of the individual units of data (called packets) that a message is divided into for efficient routing through the Internet.

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [\[RFC2119\]](#). All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[FIDO-CTAP] Brand, C., Czeskis, A., Ehrensvärd, J. et al. Eds., "Client to Authenticator Protocol (CTAP)", <https://fidoalliance.org/specs/fido-v2.0-ps-20190130/fido-client-to-authenticator-protocol-v2.0-ps-20190130.html>

[IETF-8949] Hoffman, P., "Concise Binary Object Representation (CBOR)", <https://www.ietf.org/rfc/rfc8949.txt>

[MS-ERREF] Microsoft Corporation, "[Windows Error Codes](#)".

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

[W3C-WebAuthnPKC2] W3C, "Web Authentication: An API for accessing Public Key Credentials Level 2", <https://www.w3.org/TR/webauthn-2/>

[W3C-WebAuthnPKC3] W3C, "Web Authentication: An API for accessing Public Key Credentials Level 3", <https://www.w3.org/TR/webauthn-3/>

1.2.2 Informative References

[MSFT-WebAuthnAPIS] Microsoft Corporation, "Microsoft WebAuthn APIs", <https://github.com/microsoft/webauthn/blob/master/webauthn.h>

[MSKB-5065789] Microsoft Corporation, "September 29, 2025—KB5065789", September 2025, <https://support.microsoft.com/en-au/topic/september-29-2025-kb5065789-os-builds-26200-6725-and-26100-6725-preview-fa03ce47-cec5-4d1c-87d0-cac4195b4b4e>

1.3 Overview

The Remote Desktop Protocol: WebAuthn Virtual Channel provides a way for a user to do WebAuthn operations over the RDP protocol.

More details about WebAuthn can be found in [\[W3C-WebAuthnPKC3\]](#) and [\[W3C-WebAuthnPKC2\]](#).

The WebAuthn javascript API is handled by the browser. The browser in turn calls the system-provided APIs on Windows. The system then establishes a virtual channel to the RDP client and sends the request to it. On the RDP client side, the request is decoded and processed by talking to available authenticators via the Client-to-Authenticator protocol (CTAP) protocol. See [\[FIDO-CTAP\]](#) for more details about the CTAP protocol.

1.4 Relationship to Other Protocols

This protocol uses the [\[W3C-WebAuthnPKC3\]](#), [\[W3C-WebAuthnPKC2\]](#) and [\[FIDO-CTAP\]](#) protocols.

1.5 Prerequisites/Preconditions

The Remote Desktop Protocol: WebAuthn Virtual Channel operates only after the dynamic virtual channel transport is fully established.

This protocol is message-based. It assumes preservation of the packet as a whole and does not allow for fragmentation. Additionally, it assumes that no packets are lost.

1.6 Applicability Statement

This protocol is designed to be run within the context of a **Remote Desktop Protocol (RDP)** virtual channel established between a client and a server.

1.7 Versioning and Capability Negotiation

This protocol supports versioning and capability negotiation as part of the request. A client that supports this protocol does allow this virtual channel to be opened, and a client that does not support this protocol does not allow this virtual channel to be opened.

1.8 Vendor-Extensible Fields

This protocol also uses Win32 error codes. These values are taken from the error number space as specified in [\[MS-ERREF\]](#) section 2.2. Vendors SHOULD reuse those values with their indicated meanings. Choosing any other value runs the risk of a collision in the future.

1.9 Standards Assignments

None.

2 Messages

2.1 Transport

The protocol uses a channel named WebAuthN_Channel. This channel MUST be implemented using a reliable protocol, such as **TCP**. Messages written to this channel are assumed to be complete and to arrive in order.

2.2 Message Syntax

Requests and responses are encoded in Concise Binary Object Representation (CBOR) format. For more details about CBOR, see [\[IETF-8949\]](#). CBOR encoding is used because it is used in the CTAP protocol ([\[FIDO-CTAP\]](#)) to access security keys. Hence clients needing to use external security keys need to use CBOR encoding. Platform authenticators provide their own APIs to talk to use their implementations. Overall, the WebAuthn request from the relying party is encoded along with metadata about the operation in the request message.

This protocol has two messages, a request message and a response message. The two messages perform different operations depending on their content. Each message is a CBOR map (see [\[IETF-8949\]](#), section 3.1, "Major Types," for a description of a map). The messages themselves, depending on the type of request or response, will in turn contain additional maps.

The next two sections describe the request and response messages and their elements.

2.2.1 WebAuthN_Channel Request Message

The WebAuthN_Channel request message is a CBOR map using the following keys and values.

command (key type: text string (major type 3)): An unsigned integer (major type 0) indicating the RPC command type. This field MUST be set to one of the following values:

Value	Meaning
CTAPCBOR_RPC_COMMAND_WEB_AUTHN 5	Contains both registration and assertion request for the platform authenticator as well as security keys.
CTAPCBOR_RPC_COMMAND_IUVPAA 6	Corresponds to the WebAuthn IsUserVerifyingPlatformAuthenticatorAvailable API. See [W3C-WebAuthPKC2] , section 5.1.7.
CTAPCBOR_RPC_COMMAND_CANCEL_CUR_OP 7	Cancel the current webauthn request.
CTAPCBOR_RPC_COMMAND_API_VERSION 8	Get the platform authenticator API version. <1> Callers can use the version to identify what features are available on the OS so that caller can decide whether or not a request can be fulfilled.
CTAPCBOR_RPC_COMMAND_GET_CREDENTIAL	Get credential metadata information from the platform for a relying party. <2>

Value	Meaning
9	
CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST 12	Get authenticator metadata information for user agents. <3>

request (key type: text string (major type 3)): A byte string (major type 2) containing details about the request. The contents vary depending on the **command** type.

For CTAPCBOR_RPC_COMMAND_API_VERSION, CTAPCBOR_RPC_COMMAND_IUVPAA, CTAPCBOR_RPC_COMMAND_GET_CREDENTIALS, and CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST, this field SHOULD NOT be present.

For CTAPCBOR_RPC_COMMAND_CANCEL_CUR_OP, this field MUST contain a **GUID** representing the current operation.

For CTAPCBOR_RPC_COMMAND_WEB_AUTHN, first byte MUST contain the WebAuthn command type:

Value	Meaning
CTAPCBOR_CMD_MAKE_CREDENTIAL 0x01	This command is used to create a new credential for an account for a relying party (registration phase). This is done once per account.
CTAPCBOR_CMD_GET_ASSERTION 0x02	Used to authenticate the user and sign the client data using the key created previously during the registration phase. This is also called the authenticate phase. The command is exercised multiple times after the registration phase.

The second and following bytes MUST contain a CBOR map corresponding to the WebAuthn command type in the preceding table. See sections section [2.2.1.2](#) and section [2.2.1.3](#).

flags (key type: text string (major type 3)): An unsigned integer (major type 0) containing details about the request. The value is an exclusive or (XOR) of the following values:

Value	Meaning
CTAPCLT_U2F_FLAG 0x00020000	Set to indicate the request and response will use U2F. The provider should use the U2F device interface instead of the CTAP interface.
CTAPCLT_DUAL_FLAG 0x00040000	Set to indicate to first try CTAP messages and protocol. If CTAP fails, use U2F messages..
CTAPCLT_CLIENT_PIN_REQUIRED_FLAG 0x00100000	Set to force the use of a client pin for CTAPCBOR_CMD_MAKE_CREDENTIAL.

Value	Meaning
CTAPCLT_SELECT_CREDENTIAL_ALLOW_UV_FLAG 0x00008000	When set for a login get assertion, allows user verification (UV) get assertions to select the credential.
CTAPCLT_UV_REQUIRED_FLAG 0x00400000	Set to require user verification.
CTAPCLT_UV_PREFERRED_FLAG 0x00800000	Set to indicate user verification is preferred.
CTAPCLT_UV_NOT_REQUIRED_FLAG 0x01000000	Indicates that user verification is not required.
CTAPCLT_HMAC_SECRET_EXTENSION_FLAG 0x04000000	Set to enable the hmac-secret extension for a CTAPCBOR_CMD_MAKE_CREDENTIAL request.
CTAPCLT_FORCE_U2F_V2_FLAG 0x08000000	Set to force the U2F version 2 interface to be used.

rpId (key type: text string (major type 3)): Text string (major type 3) representing Relying Party ID for which credential metadata is being requested via CTAPCBOR_RPC_COMMAND_GET_CREDENTIALS command.

authenticatorInfoLogoRequestType (key type: text string (major type 3)): An unsigned integer (major type 0) representing the type of authenticator logo requested.

Value	Meaning
WEBAUTHN_AUTHENTICATOR_LOGO_REQUEST_TYPE_NONE 0	No Logo is requested.
WEBAUTHN_AUTHENTICATOR_LOGO_REQUEST_TYPE_LIGHT 1	Light theme logo is requested.
WEBAUTHN_AUTHENTICATOR_LOGO_REQUEST_TYPE_DARK 2	Dark theme logo is requested.
WEBAUTHN_AUTHENTICATOR_LOGO_REQUEST_TYPE_ALL 3	All theme logos are requested.

hmacSecretSaltValues (key type: text string (major type 3)): A byte string (major type 2) containing hmac-secret extension salt values.

thirdPartyPayment (key type: text string (major type 3)): A Boolean value (major type 7), if set indicates that the credential is applicable for third party payment.

clientDataJSON (key type: text string (major type 3)): A byte string (major type 2) containing Client Data JSON calculated by the user agent.

remoteWebAuthn (key type: text string (major type 3)): A Boolean value (major type 7), if set indicates that the request is remote WebAuthn.

filterHybridTransport (key type: text string (major type 3)): A Boolean value (major type 7), if set indicates that the hybrid transport to be filtered out.

timeout (key type: text string (major type 3)): An unsigned integer (major type 0) representing the timeout (in milliseconds) for the operation.

transactionid (key type: text string (major type 3)): A byte array (major type 2); a GUID that is the transaction identifier.

webAuthNPara (key type: text string (major type 3)): A CBOR map (major type 5) providing parameters for authentication. See section [2.2.1.1](#) for details of the map.

2.2.1.1 webAuthNPara Map

The **webAuthNPara** is used in the WebAuthN_Channel request message to specify the parameters to use for authentication. It has the following keys and values:

wnd (key type: text string (major type 3)): An unsigned integer (major type 0) that is the window handle for the caller.

attachment (key type: text string (major type 3)): An unsigned integer (major type 0) that indicates the authenticator applicable to this operation.

Value	Meaning
WEBAUTHN_AUTHENTICATOR_ATTACHMENT_ANY 0	Use any authenticator that can satisfy the request conditions.
WEBAUTHN_AUTHENTICATOR_ATTACHMENT_PLATFORM 1	Use the platform authenticator to satisfy the request conditions. <4>
WEBAUTHN_AUTHENTICATOR_ATTACHMENT_CROSS_PLATFORM 2	Use the cross-platform roaming authenticator, such as security keys or phones, to satisfy the request conditions.

requireResident (key type: text string (major type 3)): A true or false CBOR simple value (see [\[IETF-8949\]](#), section 3.3) indicating whether resident credential keys are required.

preferResident (key type: text string (major type 3)): A true or false CBOR simple value (see [\[IETF-8949\]](#), section 3.3) indicating whether resident credential keys are preferred.

userVerification (key type: text string (major type 3)): An unsigned integer (major type 0) that represents the verification requirements. This field MUST be set to one of the following values:

Value	Meaning
WEBAUTHN_USER_VERIFICATION_REQUIREMENT_ANY 0	User verification is not required, and any setting is acceptable to the relying party.
WEBAUTHN_USER_VERIFICATION_REQUIREMENT_REQUIRED 1	User verification is required by the relying party.
WEBAUTHN_USER_VERIFICATION_REQUIREMENT_PREFERRED 2	User verification is preferred by the relying party.
WEBAUTHN_USER_VERIFICATION_REQUIREMENT_DISCOURAGED 3	User verification is discouraged by the relying party.

attestationPreference (key type: text string (major type 3)): An unsigned integer (major type 0) indicating the preferred attestation method. This field **MUST** be set to one of the following values:

Value	Meaning
WEBAUTHN_ATTESTATION_CONVEYANCE_PREFERENCE_ANY 0	Use any attestation conveyance preference.
WEBAUTHN_ATTESTATION_CONVEYANCE_PREFERENCE_NONE 1	No preference among attestation conveyance methods.
WEBAUTHN_ATTESTATION_CONVEYANCE_PREFERENCE_INDIRECT 2	Prefer indirect attestation conveyance.
WEBAUTHN_ATTESTATION_CONVEYANCE_PREFERENCE_DIRECT 3	Prefer direct attestation conveyance.

enterpriseAttestation (key type: text string (major type 3)): An unsigned integer (major type 0) indicating the enterprise attestation to use. This field **MUST** be set to one of the following values:

Value	Meaning
WEBAUTHN_ENTERPRISE_ATTESTATION_NONE 0	Enterprise attestation is not requested by the relying party.
WEBAUTHN_ENTERPRISE_ATTESTATION_VENDOR_FACILITATED 1	Enterprise attestation is requested by the relying party and authenticator can provide it if configured with this relying party.
WEBAUTHN_ENTERPRISE_ATTESTATION_PLATFORM_MANAGED 2	Enterprise attestation is requested by the relying party and the platform (OS/browser) if configured with this relying

Value	Meaning
	party can allow such attestation.

cancellationId (key type: text string (major type 3)): A byte array (major type 2); a **GUID** that is the cancellation identifier.

credLargeBlobOperation (key type: text string (major type 3)): An unsigned integer (major type 0) indicating the kind of large blob operation RP wants to perform. This field **MUST** be set to one of the following values:

Value	Meaning
0	WEBAUTHN_CRED_LARGE_BLOB_OPERATION_NONE
1	WEBAUTHN_CRED_LARGE_BLOB_OPERATION_GET
2	WEBAUTHN_CRED_LARGE_BLOB_OPERATION_SET
3	WEBAUTHN_CRED_LARGE_BLOB_OPERATION_DELETE

credLargeBlob (key type: text string (major type 3)): Byte String. Major Type 2. Representing Large Blob.

largeBlobSupport (key type: text string (major type 3)): An unsigned integer (major type 0) indicating the kind of largeBlob support RP is requesting. This field **MUST** be set to one of the following values:

Value	Meaning
0	WEBAUTHN_LARGE_BLOB_SUPPORT_NONE
1	WEBAUTHN_LARGE_BLOB_SUPPORT_REQUIRED
2	WEBAUTHN_LARGE_BLOB_SUPPORT_PREFERRED

transportHint (key type: text string (major type 3)): A Text String (major type 3) indicating the transport RP wants system to use. This field **MUST** be set to one of following values specified in section 5.8.8 of [\[W3C-WebAuthPKC3\]](#).

2.2.1.2 CTAPCBOR_CMD_MAKE_CREDENTIAL Request

This is the CBOR map used in the WebAuthN_Channel request (section [2.2.1](#)) for the CTAPCBOR_CMD_MAKE_CREDENTIAL command. The map contains details needed for the request as defined in [\[FIDO-CTAP\]](#), section 5.1.

The map has up to seven fields indicated by numeric keys. Nearly all of the fields are themselves CBOR maps. Some fields are optional.

Key	Field
1	Client data hash.
2	Relying party information.
3	User account information.
4	Algorithm preference.
5	Exclude list (optional).
6	Extensions (optional).
7	Options (optional).

The following is a text representation of an example request:

```
{
  1: h'19EEDF1F51A140B34B316293B10C2BC0E6C53860771BB1DDA4A82E5DE94A4E7C',
  2: {
    id: "ctap.dev",
    name: "WebAuthn Test Server"
  },
  3: {
    id: h'6D696B65406578616D706C652E636F6D',
    name: "mike@example.com",
    displayName: "Mike Marlowe"
  },
  4: [
    {
      alg: -7,
      type: "public-key"
    },
    {
      alg: -257,
      type: "public-key"
    }
  ],
  5: [
    {
      id: h'1CA0E7AAAB613DA3FAC3C76366A6046E',
      type: "public-key",
      transports: 1
    },
    {
      id: h'2180DA5815A4443A91050803E93EF39A',
      type: "public-key",
      transports: 1
    }
  ],
  6: {
    credBlob: h'31323132',
    largeBlob: {
      support: "preferred"
    },
    credProtect: 2,
    largeBlobKey: true,
    minPinLength: true
  },
  7: {
    rk: true
  }
}
```

}

See [FIDO-CTAP], section 5.1, for details.

2.2.1.3 CTAPCBOR_CMD_GET_ASSERTION Request

This is the CBOR map used in the WebAuthN_Channel request (section 2.2.1) for the CTAPCBOR_CMD_GET_ASSERTION command. The map contains details needed for the request as defined in [FIDO-CTAP], section 5.2.

The map has up to five fields, indicated by numeric keys. Nearly all of the fields are themselves CBOR maps. Some fields are optional.

Key	Field
1	Relying party identifier.
2	Client data hash.
3	Credential include list (optional).
4	Extensions (optional).
5	Exclude list (optional).

The following is a text representation of an example request:

```
{
  1: "ctap.dev",
  2: h'6F43BB640674BE2C2CC33AD6F0F08E450DC035F2113D1FB38A6985E32D3CB14F',
  3: [
    {
      id:
h'1C1AEF38C820EE448C7BF0EE8512BF9A813A96B73019E028E14270EB5AD74D2383EE72AC40DDABF53D3222CEBEA
861D4',
      type: "public-key"
    },
    {
      id: h'BC365B699040F8B4819833CF2F0FB4733BD424A3B043913C8CF5527799F4373A',
      type: "public-key"
    }
  ],
  4: {
    credBlob: true
  },
  5: {
    up: true
  }
}
```

See [FIDO-CTAP], section 5.2, for details.

2.2.2 WebAuthN_Channel Response Message

The WebAuthN_Channel response message is a 32-bit value followed by bytes containing data depending on the RPC command:

hresult (unsigned integer): A 32-bit HRESULT for the RPC command. See [MS-ERREF] section 2.1.

response (byte string): Variable length string of bytes depending on the RPC command. The following are the forms the **response** takes for a given RPC command:

RPC Command	Response Type
CTAPCBOR_RPC_COMMAND_WEB_AUTHN 5	A CBOR map (major type 5). See section 2.2.2.1 .
CTAPCBOR_RPC_COMMAND_IUVPA 6	A 4-byte Boolean value in little endian format. 1 for true and 0 for false.
CTAPCBOR_RPC_COMMAND_CANCEL_CUR_OP 7	No additional bytes.
CTAPCBOR_RPC_COMMAND_API_VERSION 8	A 4-byte unsigned integer in little endian format giving the API version.
CTAPCBOR_RPC_COMMAND_GET_CREDENTIALS 9	A CBOR array containing metadata information about the credentials.
CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST 12	A CBOR array containing authenticator metadata information.

2.2.2.1 CTAPCBOR_RPC_COMMAND_WEB_AUTHN Response Map

The WebAuthN_Channel response message is a CBOR map using the following keys and values.

deviceInfo (key type text string (major type 3)): A CBOR map (major type 5) containing information about the device. The map uses the following keys and values:

maxMsgSize (key type text string (major type 3)): An unsigned 32-bit integer (major type 0) providing the maximum message size the authenticator can process.

maxSerializedLargeBlobArray (key type text string (major type 3)): An unsigned 32-bit integer (major type 0) providing the maximum size serialized large blob the authenticator can process.

providerType (key type text string (major type 3)): A text string (major type 3) representing the provider type. Used only for information purposes. One of the following values:

Value	Meaning
CTAPHID_PROVIDER_TYPE L"Hid"	An Hid provider.

Value	Meaning
CTAPNFC_PROVIDER_TYPE L"Nfc"	An Nfc provider.
CTAPBLE_PROVIDER_TYPE L"Ble"	A Ble provider.
WEBAUTHN_PLATFORM_PROVIDER_TYPE L"Platform"	A platform provider.

providerName (key type text string (major type 3)): A text string (major type 3) representing the provider's name. Used only for information.

devicePath (key type text string (major type 3)): A text string (major type 3) providing the path to the authenticator.

Manufacturer (key type text string (major type 3)): A text string (major type 3) representing the manufacturer of the authenticator.

Product (key type text string (major type 3)): A text string (major type 3) representing the product name of the authenticator.

aaGuid (key type text string (major type 3)): A 16 byte byte-string (major type 2) containing the Authenticator Attestation GUID (AAGUID) for the authenticator.

residentKey (key type text string (major type 3)): A true or false CBOR simple value that indicates whether a resident credential was created.

uvStatus (key type text string (major type 3)): An unsigned integer (major type 0) giving the verification status of the operation. This field **MUST** be set to one of the following values:

Value	Meaning
WEBAUTHN_USER_VERIFICATION_REQUIREMENT_ANY 0	Use any verification requirement.
WEBAUTHN_USER_VERIFICATION_REQUIREMENT_REQUIRED 1	A verification is required.
WEBAUTHN_USER_VERIFICATION_REQUIREMENT_PREFERRED 2	A verification is preferred.
WEBAUTHN_USER_VERIFICATION_REQUIREMENT_DISCOURAGED 3	Verification is discouraged.

uvRetries (key type text string (major type 3)): An unsigned integer (major type 0) representing the number of verification retries available for the authenticator.

credWithHmacSecretArray (key type text string (major type 3)): A byte string (major type 2) containing hmacSecret extension response value.

thirdPartyPayment (key type: text string (major type 3)): A Boolean value (major type 7) representing whether the credential created is applicable for third party payment.

transports (key type: text string (major type 3)): An unsigned integer (major type 0) representing the transports supported by the credential.

Status (key type text string (major type 3)): An unsigned integer (major type 0) giving the status of the overall operation.

Response (key type text string (major type 3)): Contains the response to the individual operation corresponding to the individual WebAuthn command. The response consists of a single byte indicating success or failure. A value of 0x00 indicates success. Any other value is an error. See [\[FIDO-CTAP\]](#), section 6.3 for values and meaning. The single-byte code is followed by a CBOR map containing details of the response. See section [2.2.2.1.1](#) and section [2.2.2.1.2](#) for the response maps.

2.2.2.1.1 CTAP MakeCredential Response

This is the CBOR map used in the CTAPCBOR_RPC_COMMAND_WEB_AUTHN response map for a response to a request to make a credential (see section [2.2.2.1](#)).

See [\[FIDO-CTAP\]](#), section 6.2, for details of the map keys, values, and data types.

The following is a text representation of an example response:

```
{
  1: "packed",
  2:
  h'3D3CA4E1D7E11F604B32D0EFE18B449B057CC736ECE0BF820A42555426DB8834C500000003D8522D9F575B48668
  8A9BA99FA02F35B00301C1AEF38C820EE448C7BF0EE8512BF9A813A96B73019E028E14270EB5AD74D2383EE72AC40
  DDABF53D3222CEBEA861D4A50102032620012158201C1AEF38C820EE448C7BF0EE85912D0C4650F2A7B830431CBC1
  3F49714D11FBA2258206A06795A704BD761068E13875D928FCDCC6C4E7F2AF4BCEC849098BCCF84F72DA368637265
  64426C6F62F56B6372656450726F74656374036B686D61632D736563726574F5',
  3: {
    alg: -7,
    sig:
    h'3045022100D8233DF9A938527864729037AD07DF2294F7CE764C886EF1EDC2F3AB1BD9CF6802201282A88B41A53
    FB2B338CBFC4FFF713A0DAA097BA359E1E72B60D93983716E8F',
    x5c: [
      h'308202D8308201C0A003020102020900FF876C2DAF7379C8300D06092A864886F70D01010B0500302E312C302A0
      603550403132359756269636F2055324620526F6F742043412053657269616C203435373230303633313020170D31
      34303830313030303030305A180F32303530303930343030303030305A306E310B300906035504061302534531123
      010060355040A0C0959756269636F20414231223020060355040B0C1941757468656E74696361746F722041747465
      73746174696F6E3127302506035504030C1E59756269636F205532462045452053657269616C20373632303837343
      2333059301306072A8648CE3D020106082A8648CE3D0301070342000425F123A048283FC5796CCF887D99489FD935
      C24198C4B5D8D5B2C2BFD7DD5D15AFE45B7070776567D5B5B0B23E04560B5BEA77B483B1F6491E53A3F2BEE6A39AA
      38181307F3013060A2B0601040182C40A0D0104050403050506302206092B0601040182C40A020415312E332E362E
      312E342E312E34313438322E312E393013060B2B0601040182E51C0201010404030205203021060B2B0601040182E
      51C01010404120410D8522D9F575B486688A9BA99FA02F35B300C0603551D130101FF04023000300D06092A864886
      F70D01010B0500038201010052B06949DBAAD1A64C1BA9EBEC198B317EC31F9A37363BA5161B342E3A49CAD504F34E
      7428BB896E9CFD28D03AD10CE325A06838E9B6C4ECB17AD40D090A16C9E7C34498332FF853B62747E8FCDF00DAE62
      756E57BD40B16D677907A835C0435A2EBCE9B0B9069CA122BF9D964A73206AF74FF3C00144EBFF3DE7C7758D3147C
      8C2F9FE87C12F2A9675A2046B01076361A99721871FA78FB0DE2945B579F9166C48AD2FD50C3CE56C8221A75083F6
      56119394368FF17D2C920C63A09F01ED2501146B7DF1AB3970A2A32938FA9A517AF471085E160B3CA79764231746B
      A6ABBA68E0D13CE259796BCD2A03AD83C74E15331328EAB438E6A4197CB12EC6FD1E388'
    ]
  },
  5: h'523FC0E9385A21D9A9A427B494014DDF3F9934838C05CD4C97BDF564FC3C8AC8'
}
```

2.2.2.1.2 CTAP GetAssertion Response

This is the CBOR map used in the CTAPCBOR_RPC_COMMAND_WEB_AUTHN response map for a response to a request to get an assertion (see section [2.2.2.1](#)).

See [\[FIDO-CTAP\]](#), section 6.2, for details of the map keys, values, and data types.

The following is a text representation of an example response:

```
{
  1: {
    id:
    h'1C1AEF38C820EE448C7BF0EE8512BF9A813A96B73019E028E14270EB5AD74D2383EE72AC40DDABF53D3222CEBEA
    861D4',
    type: "public-key"
  },
  2:
  h'3D3CA4E1D7E11F604B32D0EFE18B449B057CC736ECE0BF820A42555426DB88348500000006A26863726564426C6
  F6244313231326B686D61632D736563726574585051144A81FFFCED51BB81932851ABB84D17325C764C95C5016B5E
  D0F80FA763728F7D410B39103FC502E115D7DAA326D6D0FA59A06BB20DFB4DA9AE31F1E3840D520FF29A01CD4B98D
  7B555730C289673',
  3:
  h'3045022033626FC0675341C2EBC5D5A049D739F517110F749685E12C3DABCB2EB129E03022100D3CE29E77F8EB
  B3A0B63D858E56ACE6B3A8702BF5113BBE7A081D352F844914F',
  4: {
    id: h'6D696B65406578616D706C652E636F6D'
  }
}
```

2.2.2.2 CTAPCBOR_RPC_COMMAND_GET_CREDENTIALS Response Map

This is a CBOR array which contains metadata about the credentials known to the platform for the requested relying party

Individual credential metadata information is represented in a CBOR map with following fields.

Key (Unsigned Integer – CBOR Major Type 0)	Value
0	Version Information (Unsigned Integer. CBOR Major Type 0).
1	Credential ID (Byte String. Major Type 2).
2	RP Entity Information.
3	User Entity Information.
4	Removable or not from the Authenticator.
5	Backed-up Credential or not. (Boolean. CBOR Major Type 7).
6	Authenticator Name. (String. Major Type

Key (Unsigned Integer – CBOR Major Type 0)	Value
	3).
7	Authenticator Logo.
8	Third Party Payment (Boolean. CBOR Major Type 7).
9	Transports Supported. (Unsigned Integer. CBOR Major Type 0).

The following is a text representation of an example response:

```
[
{
  0: 4,
  1: h'1C337E7A9F27200C98BE1E0337A501C02CBFA7215A85EA8CD9E187A1E97EFDDDD',
  2: {
    id: "ctap.dev",
    name: "WebAuthn Test Server"
  },
  3: {
    id: h'626F62406578616D706C652E636F6D',
    name: "bob@example.com",
    displayName: "Bob Smith"
  },
  4: true,
  5: false,
  6: "Windows Hello",
  7:
h'3C7376672076696577426F783D22302030203339203339222066696C6C3D226E6F6E652220786D6C6E733D2
2687474703A2F2F7777772E77332E6F72672F323030302F737667223E3C726563742077696474683D22343922
206865696768743D223439222072783D2232222066696C6C3D227768697465222F3E3C6D61736B2069643D226
D61736B305F31313738315F31363230313422207374796C653D226D61736B2D747970653A6C756D696E616E63
6522206D61736B556E6974733D227573657253706163654F6E5573652220783D22382220793D2238222077696
474683D22333322206865696768743D223333223E3C7061746820643D224D3431203848385634314834315638
5A222066696C6C3D227768697465222F3E3C2F6D61736B3E3C67206D61736B3D2275726C28236D61736B305F3
1313738315F31363230313422207472616E73666F726D3D227472616E736C617465282D352E352C2D352922
3E3C706174682066696C6C2D72756C653D226576656E6F64642220636C69702D72756C653D226576656E6F646
42220643D224D31372E3632352031362E32354331372E3632352031322E383332372032302E33393532203130
2E303632352032332E383132352031302E303632354332362E393137342031302E303632352032392E3438383
32031322E333439352032392E393332312031352E333330384333302E3533352031342E393734342033312E31
3837322031342E363932352033312E383736312031342E343937384333312E303732372031302E37383331203
2372E3736373620382032332E3831323520384331392E3235363120382031352E353632352031312E36393336
2031352E353632352031362E32354331352E353632352032302E383036342031392E323536312032342E35203
2332E383132352032342E354332342E353932312032342E352032352E333436352032342E333931392032362E
303631352032342E313839374332352E393339332032332E3632352032352E3837352032332E3033383820323
52E3837352032322E343337354332352E3837352032322E333138372032352E383737352032322E3230303420
32352E383832352032322E303832384332352E32333532032322E333132352032342E353338352032322E343
337352032332E383132352032322E343337354332302E333935322032322E343337352031372E363235203139
2E363637332031372E3632352031362E32355A4D31332E352032362E353632354832362E393738374332372E3
43237382032372E333338382032372E393939382032382E303335322032382E3636382032382E363235483133
2E354331322E3935332032382E3632352031322E343238342032382E383432342031322E303431362032392E3
23239314331312E363534382032392E363135382031312E343337352033302E313430352031312E3433373520
33302E363837354331312E343337352033322E3934382031322E333333372033342E393538332031342E32353
6322033362E343331384331362E323038322033372E393237392031392E333130342033382E39333735203233
2E383132352033382E393337354332362E323832372033382E393337352032382E333331342033382E3633333
52033302033382E313132315634302E323630324332382E323230392034302E373432362032362E3135383620
34312032332E383132352034314331392E303333342034312031352E343332342033392E393331382031332E3
03031362033382E303638384331302E353431332033362E3138333220392E3337352033332E3535323920392E
```

```

3337352033302E3638373543392E3337352032392E35393333520392E38303935392032382E353434332031302
E353833322032372E373730374331312E333536382032362E393937312031322E3430362032362E3536323520
31332E352032362E353632355A4D32382E323839362032342E34394332382E303631372032332E38343537203
2372E393337352032332E313532332032372E393337352032322E34334332372E393337352031392E30313639
2033302E373038332031362E32352033342E313235362031362E32354333372E353432382031362E323520343
02E333133312031392E303136392034302E333133312032322E34334334302E333133312032352E3138382033
382E353034312032372E353234312033362E303036362032382E333139324C33372E393433372033302E30303
4324333382E333438342033302E333536312033382E333438342033302E393833392033372E39343337203331
2E333335374C33352E313536392033332E37364C33372E393335372033362E3233344333382E3331323320333
62E353639332033382E333331392033372E313531322033372E393738382033372E353131314C33342E383134
312034302E373335324333342E343436332034312E313039392033332E383334342034312E303833332033332
E353030352034302E363738324C33322E323634362033392E313738364333322E313334332033392E30323036
2033322E303633312033382E383232322033322E303633312033382E363137344C33322E303632352033322E3
33531335632382E323538344333322E303034332032382E323337382033312E393436362032382E3231363220
33312E383839332032382E3139344333302E323133332032372E353434382032382E383930342032362E31383
7372032382E323839362032342E34395A4D33342E313235362032322E34334333352E323634372032322E3433
2033362E313838312032312E353037372033362E313838312032302E33374333362E313838312031392E32333
232033352E323634372031382E33312033342E313235362031382E33314333322E393836352031382E333120
33322E303633312031392E323332332033322E303633312032302E33374333322E303633312032312E3530373
72033322E393836352032322E34332033342E313235362032322E343335A222066696C6C3D22626C61636B2220
66696C6C2D6F7061636974793D22302E38393536222F3E3C2F673E3C2F7376673E00',
      8: false,
      9: 16
    }
  ]
]

```

2.2.2.3 CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST Response Map

This is a CBOR array which contains metadata about the authenticator known to the platform for user agents

Individual authenticator metadata information is represented in a CBOR map with following fields.

Key (Unsigned Integer – CBOR Major Type 0)	Value
1	Version Information (Unsigned Integer. CBOR Major Type 0).
2	Authenticator ID (Byte String. Major Type 2).
3	Authenticator Name (String. Major Type 3).
4	Authenticator Logo (Byte String. Major Type 2).
5	IsLocked. (Boolean. CBOR Major Type 7).

The following is a text representation of an example response:

```

[
  {
    1: 1,

```

```
2: h'08987058CADC4B81B6E130DE50DCBE96',
3: "Windows Hello",
4:
h'3C7376672069643D224C617965725F312220786D6C6E733D22687474703A2F2F7777772E77332E6F72672F3
23030302F737667222076696577426F783D223020302032353620323536223E3C646566733E3C7374796C653E
2E636C732D317B66696C6C3A233030373864343B7374726F6B652D77696474683A3070783B7D3C2F7374796C6
53E3C2F646566733E3C7265637420636C6173733D22636C732D312220783D2232342E32352220793D2232342E
3235222077696474683D2239382E333522206865696768743D2239382E3335222F3E3C7265637420636C61737
33D22636C732D312220783D223133332E342220793D2232342E3235222077696474683D2239382E3335222068
65696768743D2239382E3335222F3E3C7265637420636C6173733D22636C732D312220783D2232342E3235222
0793D223133332E34222077696474683D2239382E333522206865696768743D2239382E3335222F3E3C726563
7420636C6173733D22636C732D312220783D223133332E342220793D223133332E34222077696474683D22393
82E333522206865696768743D2239382E3335222F3E3C2F7376673E',
5: false
}
]
```

3 Protocol Details

3.1 Client and Server Details

The protocol has two main operations: MakeCredential (section [2.2.1.2](#)) and GetAssertion (section [2.2.1.3](#)). The MakeCredential operation registers a credential on the authenticator for the replying party. The GetAssertion operation authenticates the user to the relying party using the authenticator.

This protocol is designed to be closer to the WebAuthn layer rather than the CTAP layer. Individual fields in the request map to the WebAuthn options defined at the WebAuthn layer.

Once a request is received by the client, the client determines which authenticator can satisfy the request. The process of determining which authenticator supports particular capabilities and whether it can satisfy a request are defined in the CTAP specification. See [\[FIDO-CTAP\]](#), sections 4 and 5.4.

3.1.1 Abstract Data Model

None.

3.1.2 Timers

None.

3.1.3 Initialization

None.

3.1.4 Higher-Layer Triggered Events

None.

3.1.5 Message Processing Events and Sequencing Rules

None.

3.1.6 Timer Events

None.

3.1.7 Other Local Events

None.

4 Protocol Examples

The following sections provide examples of requests and responses for different commands.

4.1 CTAPCBOR_RPC_COMMAND_API_VERSION

4.1.1 Request

Complete request (a CBOR map):

```
A467636F6D6D616E640565666C616773006774696D656F7574006D7472616E73616374696F6E4964500000000000  
00000000000000000000
```

Textual representation of the CBOR encoding:

```
{"command": 5, "flags": 0, "timeout": 0, "transactionId":  
h'00000000000000000000000000000000'}
```

4.1.2 Response

Full response including HRESULT and API Version:

```
0x00000004
```

The preceding response indicates a successful call (0x00000000 as first 32 bits) and that the API version is 4.

4.2 CTAPCBOR_RPC_COMMAND_IUVPAA

4.2.1 Request

Full request (a CBOR map):

```
0xA467636F6D6D616E640665666C616773006774696D656F7574006D7472616E73616374696F6E49645000000000  
00000000000000000000
```

Textual representation of the CBOR encoding:

```
{"command": 6, "flags": 0, "timeout": 0, "transactionId":  
h'00000000000000000000000000000000'}
```

4.2.2 Response

Full response including HRESULT and IUVPAA:

```
0x00000001
```



```

    timeout: 300000,
    transactionId: h'26C98F33FCD61D4DBA5B27570F8293DF',
    request:
h'01A701582019EEDF1F51A140B34B316293B10C2BC0E6C53860771BB1DDA4A82E5DE94A4E7C02A26269646863746
1702E646576646E616D6574576562417574686E20546573742053657276657203A3626964506D696B65406578616D
706C652E636F6D646E616D65706D696B65406578616D706C652E636F6D6B646973706C61794E616D656C4D696B652
04D61726C6F77650482A263616C672664747970656A7075626C69632D6B6579A263616C6739010064747970656A70
75626C69632D6B65790582A3626964501CA0E7AAAB613DA3FAC3C76366A6046E64747970656A7075626C69632D6B6
5796A7472616E73706F72747301A3626964502180DA5815A4443A91050803E93EF39A64747970656A7075626C6963
2D6B65796A7472616E73706F7274730106A56863726564426C6F624431323132696C61726765426C6F62A16773757
0706F7274697072656665727265646B6372656450726F74656374026C6C61726765426C6F624B6579F56C6D696E50
696E4C656E677468F507A162726BF5',
    webAuthNPara: {
      wnd: 197430,
      attachment: 2,
      requireResident: true,
      preferResident: false,
      userVerification: 1,
      attestationPreference: 3,
      enterpriseAttestation: 0,
      cancellationId: h'DAED4B74000000000000000000000000',
      largeBlobSupport: 2
    },
    filterHybridTransport: false,
    thirdPartyPayment: false,
    clientDataJSON:
h'7B2274797065223A22776562617574686E2E637265617465222C226368616C6C656E6765223A226158774C66703
935524A6D5F5F366E4D753442504C7152476D42453675504D75582D3170506B6542517841222C226F726967696E22
3A2268747470733A2F2F637461702E646576222C2263726F73734F726967696E223A66616C73652C226F746865725
F6B6579735F63616E5F62655F61646465645F68657265223A22646F206E6F7420636F6D7061726520636C69656E74
446174614A534F4E20616761696E737420612074656D706C6174652E205365652068747470733A2F2F676F6F2E676
C2F796162506578227D'
  }
}

```

Inner MakeCredential request details after the command type(0x01):

```

A701582019EEDF1F51A140B34B316293B10C2BC0E6C53860771BB1DDA4A82E5DE94A4E7C02A262696468637461702
E646576646E616D6574576562417574686E20546573742053657276657203A3626964506D696B65406578616D706C
652E636F6D646E616D65706D696B65406578616D706C652E636F6D6B646973706C61794E616D656C4D696B65204D6
1726C6F77650482A263616C672664747970656A7075626C69632D6B6579A263616C6739010064747970656A707562
6C69632D6B65790582A3626964501CA0E7AAAB613DA3FAC3C76366A6046E64747970656A7075626C69632D6B65796
A7472616E73706F72747301A3626964502180DA5815A4443A91050803E93EF39A64747970656A7075626C69632D6B
65796A7472616E73706F7274730106A56863726564426C6F624431323132696C61726765426C6F62A167737570706
F7274697072656665727265646B6372656450726F74656374026C6C61726765426C6F624B6579F56C6D696E50696E
4C656E677468F507A162726BF5

```

Textual representation of the inner map:

```

{
  1: h'19EEDF1F51A140B34B316293B10C2BC0E6C53860771BB1DDA4A82E5DE94A4E7C',
  2: {
    id: "ctap.dev",
    name: "WebAuthn Test Server"
  },
  3: {
    id: h'6D696B65406578616D706C652E636F6D',
    name: "mike@example.com",
    displayName: "Mike Marlowe"
  },
  4: [
    {
      alg: -7,
      type: "public-key"
    }
  ]
}

```

```

      alg: -257,
      type: "public-key"
    },
  ],
  5: [
    {
      id: h'1CA0E7AAAB613DA3FAC3C76366A6046E',
      type: "public-key",
      transports: 1
    },
    {
      id: h'2180DA5815A4443A91050803E93EF39A',
      type: "public-key",
      transports: 1
    }
  ],
  6: {
    credBlob: h'31323132',
    largeBlob: {
      support: "preferred"
    },
    credProtect: 2,
    largeBlobKey: true,
    minPinLength: true
  },
  7: {
    rk: true
  }
}

```

4.4.1.2 Response

Full RPC response including the HRESULT and the CBOR map:

```

0x0000A36A646576696365496E666FAE6A6D61784D736753697A651904B0781B6D617853657269616C697A65644C6
1726765426C6F6241727261791904006C70726F766964657254797065634869646C70726F76696465724E616D6578
184D6963726F736F66744374617048696450726F76696465726A6465766963655061746878505C5C3F5C686964237
669645F31303530267069645F3034303223382639336334326626302630303030237B34643165353562322D663136
662D313163662D383863622D3030313131313030303033307D6C6D616E7566616374757265726659756269636F677
0726F647563746C597562694B6579204649444F66616147756964509F2D52D85B57664888A9BA99FA02F35B6B7265
736964656E744B6579F5781A63726564656E7469616C4C697374496E646578506C75734F6E6520687576537461747
57301697576526574726965730371746869726450617274795061796D656E74F46A7472616E73706F727473016673
74617475730068726573706F6E736559044100A401667061636B65640258D93D3CA4E1D7E11F604B32D0EFE18B449
B057CC736ECE0BF820A42555426DB8834C500000003D8522D9F575B486688A9BA99FA02F35B00301C1AEF38C820EE
448C7BF0EE8512BF9A813A96B73019E028E14270EB5AD74D2383EE72AC40DDABF53D3222CEBEA861D4A5010203262
0012158201C1AEF38C820EE448C7BF0EE85912D0C4650F2A7B830431CBC13F49714D11FBA2258206A06795A704BD7
61068E13875D928FCDC6C4E7F2AF4BCEC849098BCCF84F72DA36863726564426C6F62F56B6372656450726F74656
374036B686D61632D736563726574F503A363616C67266373696758473045022100D8233DF9A938527864729037AD
07DF2294F7CE764C886EF1EDC2F3AB1BD9CF6802201282A88B41A53FB2B338CBFC4FFF713A0DAA097BA359E1E72B6
0D93983716E8F63783563815902DC308202D8308201C0A003020102020900FF876C2DAF7379C8300D06092A864886
F70D01010B0500302E312C302A0603550403132359756269636F2055324620526F6F742043412053657269616C203
435373230303633313020170D3134303830313030303030305A180F32303530303930343030303030305A306E310B
300906035504061302534531123010060355040A0C0959756269636F20414231223020060355040B0C19417574686
56E74696361746F72204174746573746174696F6E3127302506035504030C1E59756269636F205532462045452053
657269616C203736323038373432333059301306072A8648CE3D020106082A8648CE3D0301070342000425F123A04
8283FC5796CCF887D99489FD935C24198C4B5D8D5B2C2BFD7DD5D15AFE45B7070776567D5B5B0B23E04560B5BEA77
B483B1F6491E53A3F2BEE6A39AA38181307F3013060A2B0601040182C40A0D0104050403050506302206092B06010
40182C40A020415312E332E362E312E342E312E34313438322E312E393013060B2B0601040182E51C020101040403
0205203021060B2B0601040182E51C01010404120410D8522D9F575B486688A9BA99FA02F35B300C0603551D13010
1FF04023000300D06092A864886F70D01010B0500038201010052B06949DBAAD1A64C1BA9EBC198B317EC31F9A373
63BA5161B342E3A49CAD504F34E7428BB896E9CFD28D03AD10CE325A06838E9B6C4ECB17AD40D090A16C9E7C34498
332FF853B62747E8FCDF00DAE62756E57BD40B16D677907A835C0435A2EBCE9B0B9069CA122BF9D964A73206AF74F
F3C00144EBFF3DE7C7758D3147C8C2F9FE87C12F2A9675A2046B01076361A99721871FA78FB0DE2945B579F9166C4
8AD2FD50C3CE56C8221A75083F656119394368FF17D2C920C63A09F01ED2501146B7DF1AB3970A2A32938FA9A517A

```

F471085E160B3CA79764231746BA6ABBA68E0D13CE259796BCD2A03AD83C74E15331328EAB438E6A4197CB12EC6FD
1E388055820523FC0E9385A21D9A9A427B494014DDF3F9934838C05CD4C97BDF564FC3C8AC8

Textual representation of the response following the HRESULT:

```
{
  deviceInfo: {
    maxMsgSize: 1200,
    maxSerializedLargeBlobArray: 1024,
    providerType: "Hid",
    providerName: "MicrosoftCtapHidProvider",
    devicePath: "\\.\\" + "\\hid#vid_1050&pid_0402#8&93c42f&0&0000#{4d1e55b2-f16f-11cf-88cb-001111000030}",
    manufacturer: "Yubico",
    product: "YubiKey FIDO",
    aaGuid: h'9F2D52D85B57664888A9BA99FA02F35B',
    residentKey: true,
    credentialListIndexPlusOne: -1,
    uvStatus: 1,
    uvRetries: 3,
    thirdPartyPayment: false,
    transports: 1
  },
  status: 0,
  response:
h'00A401667061636B65640258D93D3CA4E1D7E11F604B32D0EFE18B449B057CC736ECE0BF820A42555426DB8834C
500000003D8522D9F575B486688A9BA99FA02F35B00301C1AEF38C820EE448C7BF0EE8512BF9A813A96B73019E028
E14270EB5AD74D2383EE72AC40DDABF53D3222CEBEA861D4A50102032620012158201C1AEF38C820EE448C7BF0EE8
5912D0C4650F2A7B830431CBC13F49714D11FBA2258206A06795A704BD761068E13875D928FCDCC6C4E7F2AF4BCEC
849098BCCF84F72DA36863726564426C6F62F56B6372656450726F74656374036B686D61632D736563726574F503A
363616C67266373696758473045022100D8233DF9A938527864729037AD07DF2294F7CE764C886EF1EDC2F3AB1BD9
CF6802201282A88B41A53FB2B338CBFC4FFF713A0DAA097BA359E1E72B60D93983716E8F63783563815902DC30820
2D8308201C0A003020102020900FF876C2DAF7379C8300D06092A864886F70D01010B0500302E312C302A06035504
03132359756269636F2055324620526F6F742043412053657269616C203435373230303633313020170D313430383
0313030303030305A180F32303530303930343030303030305A306E310B3009060355040613025345311230100603
55040A0C0959756269636F20414231223020060355040B0C1941757468656E74696361746F7220417474657374617
4696F6E3127302506035504030C1E59756269636F205532462045452053657269616C203736323038373432333059
301306072A8648CE3D020106082A8648CE3D0301070342000425F123A048283FC5796CCF887D99489FD935C24198C
4B5D8D5B2C2BFD7DD5D15AFE45B7070776567D5B5B0B23E04560B5BEA77B483B1F6491E53A3F2BEE6A39AA3818130
7F3013060A2B0601040182C40A0D0104050403050506302206092B0601040182C40A020415312E332E362E312E342
E312E3431343832E312E393013060B2B0601040182E51C0201010404030205203021060B2B0601040182E51C0101
0404120410D8522D9F575B486688A9BA99FA02F35B300C0603551D130101FF04023000300D06092A864886F70D010
10B0500038201010052B06949DBAAD1A64C1BA9EBC198B317EC31F9A37363BA5161B342E3A49CAD504F34E7428BB8
96E9CFD28D03AD10CE325A06838E9B6C4ECB17AD40D090A16C9E7C34498332FF853B62747E8FCDF00DAE62756E57B
D40B16D677907A835C0435A2EBCE9B0B9069CA122BF9D964A73206AF74FF3C00144EBFF3DE7C7758D3147C8C2F9FE
87C12F2A9675A2046B01076361A99721871FA78FB0DE2945B579F9166C48AD2FD50C3CE56C8221A75083F65611939
4368FF17D2C920C63A09F01ED2501146B7DF1AB3970A2A32938FA9A517AF471085E160B3CA79764231746BA6ABBA6
8E0D13CE259796BCD2A03AD83C74E15331328EAB438E6A4197CB12EC6FD1E388055820523FC0E9385A21D9A9A427B
494014DDF3F9934838C05CD4C97BDF564FC3C8AC8'
}
```

Inner Authenticator response details after first byte indicating success(0x00):

A401667061636B65640258D93D3CA4E1D7E11F604B32D0EFE18B449B057CC736ECE0BF820A42555426DB8834C5000
00003D8522D9F575B486688A9BA99FA02F35B00301C1AEF38C820EE448C7BF0EE8512BF9A813A96B73019E028E142
70EB5AD74D2383EE72AC40DDABF53D3222CEBEA861D4A50102032620012158201C1AEF38C820EE448C7BF0EE85912
D0C4650F2A7B830431CBC13F49714D11FBA2258206A06795A704BD761068E13875D928FCDCC6C4E7F2AF4BCEC8490
98BCCF84F72DA36863726564426C6F62F56B6372656450726F74656374036B686D61632D736563726574F503A3636
16C67266373696758473045022100D8233DF9A938527864729037AD07DF2294F7CE764C886EF1EDC2F3AB1BD9CF68
02201282A88B41A53FB2B338CBFC4FFF713A0DAA097BA359E1E72B60D93983716E8F63783563815902DC308202D83
08201C0A003020102020900FF876C2DAF7379C8300D06092A864886F70D01010B0500302E312C302A060355040313
2359756269636F2055324620526F6F742043412053657269616C203435373230303633313020170D3134303830313
030303030305A180F32303530303930343030303030305A306E310B30090603550406130253453112301006035504
0A0C0959756269636F20414231223020060355040B0C1941757468656E74696361746F72204174746573746174696
F6E3127302506035504030C1E59756269636F205532462045452053657269616C2037363230383734323330593013

06072A8648CE3D020106082A8648CE3D0301070342000425F123A048283FC5796CCF887D99489FD935C24198C4B5D
8D5B2C2BFD7DD5D15AFE45B7070776567D5B5B0B23E04560B5BEA77B483B1F6491E53A3F2BEE6A39AA38181307F30
13060A2B0601040182C40A0D0104050403050506302206092B0601040182C40A020415312E332E362E312E342E312
E34313438322E312E393013060B2B0601040182E51C0201010404030205203021060B2B0601040182E51C01010404
120410D8522D9F575B486688A9BA99FA02F35B300C0603551D130101FF04023000300D06092A864886F70D01010B0
500038201010052B06949DBAAD1A64C1BA9EBC198B317EC31F9A37363BA5161B342E3A49CAD504F34E7428BB896E9
CFD28D03AD10CE325A06838E9B6C4ECB17AD40D090A16C9E7C34498332FF853B62747E8FCDF00DAE62756E57BD40B
16D677907A835C0435A2EBCE9B0B9069CA122BF9D964A73206AF74FF3C00144EBFF3DE7C7758D3147C8C2F9FE87C1
2F2A9675A2046B01076361A99721871FA78FB0DE2945B579F9166C48AD2FD50C3CE56C8221A75083F656119394368
FF17D2C920C63A09F01ED2501146B7DF1AB3970A2A32938FA9A517AF471085E160B3CA79764231746BA6ABBA68E0D
13CE259796BCD2A03AD83C74E15331328EAB438E6A4197CB12EC6FD1E388055820523FC0E9385A21D9A9A427B4940
14DDF3F9934838C05CD4C97BDF564FC3C8AC8

Textual representation of the inner Authenticator response:

```
{
  1: "packed",
  2:
  h'3D3CA4E1D7E11F604B32D0EFE18B449B057CC736ECE0BF820A42555426DB8834C500000003D8522D9F575B4
  86688A9BA99FA02F35B00301C1AEF38C820EE448C7BF0EE8512BF9A813A96B73019E028E14270EB5AD74D2383
  EE72AC40DDABF53D322CEBEA861D4A50102032620012158201C1AEF38C820EE448C7BF0EE85912D0C4650F2A
  7B830431CBC13F49714D11FBA2258206A06795A704BD761068E13875D928FCDCC6C4E7F2AF4BCEC849098BCCF
  84F72DA36863726564426C6F62F56B6372656450726F74656374036B686D61632D736563726574F5',
  3: {
    alg: -7,
    sig:
    h'3045022100D8233DF9A938527864729037AD07DF2294F7CE764C886EF1EDC2F3AB1BD9CF6802201282A88B4
    1A53FB2B338CBFC4FFF713A0DAA097BA359E1E72B60D93983716E8F',
    x5c: [

      h'308202D8308201C0A003020102020900FF876C2DAF7379C8300D06092A864886F70D01010B0500302E312C3
      02A0603550403132359756269636F2055324620526F6F742043412053657269616C2034353732303036333130
      20170D3134303830313030303030305A180F32303530303930343030303030305A306E310B300906035504061
      302534531123010060355040A0C0959756269636F20414231223020060355040B0C1941757468656E74696361
      746F72204174746573746174696F6E3127302506035504030C1E59756269636F2055324620454520536572696
      16C203736323038373432333059301306072A8648CE3D020106082A8648CE3D0301070342000425F123A04828
      3FC5796CCF887D99489FD935C24198C4B5D8D5B2C2BFD7DD5D15AFE45B7070776567D5B5B0B23E04560B5BEA7
      7B483B1F6491E53A3F2BEE6A39AA38181307F3013060A2B0601040182C40A0D0104050403050506302206092B
      0601040182C40A020415312E332E362E312E342E312E34313438322E312E393013060B2B0601040182E51C020
      1010404030205203021060B2B0601040182E51C01010404120410D8522D9F575B486688A9BA99FA02F35B300C
      0603551D130101FF04023000300D06092A864886F70D01010B0500038201010052B06949DBAAD1A64C1BA9EBC
      198B317EC31F9A37363BA5161B342E3A49CAD504F34E7428BB896E9CFD28D03AD10CE325A06838E9B6C4ECB17
      AD40D090A16C9E7C34498332FF853B62747E8FCDF00DAE62756E57BD40B16D677907A835C0435A2EBCE9B0B90
      69CA122BF9D964A73206AF74FF3C00144EBFF3DE7C7758D3147C8C2F9FE87C12F2A9675A2046B01076361A997
      21871FA78FB0DE2945B579F9166C48AD2FD50C3CE56C8221A75083F656119394368FF17D2C920C63A09F01ED2
      501146B7DF1AB3970A2A32938FA9A517AF471085E160B3CA79764231746BA6ABBA68E0D13CE259796BCD2A03A
      D83C74E15331328EAB438E6A4197CB12EC6FD1E388'
    ]
  },
  5: h'523FC0E9385A21D9A9A427B494014DDF3F9934838C05CD4C97BDF564FC3C8AC8'
}
```

4.4.2 CTAPCBOR_CMD_GET_ASSERTION

4.4.2.1 Request

Full request, a CBOR map:

A967636F6D6D616E640565666C6167731A004000006774696D656F75741A000493E06D7472616E73616374696F6E4
964509C6904B32DBF974E96254E9B65D0AF8E677265717565737458BF02A50168637461702E6465760258206F43BB
640674BE2C2CC33AD6F0F08E450DC035F2113D1FB38A6985E32D3CB14F0382A262696458301C1AEF38C820EE448C7

Textual representation of the request:

Inner GetAssertion request details following the command type(0x02):

Textual representation of the preceding GetAssertion request:


```
response:
h'00A401A262696458301C1AEF38C820EE448C7BF0EE8512BF9A813A96B73019E028E14270EB5AD74D2383EE72AC4
0DDABF53D3222CEBEA861D464747970656A7075626C69632D6B65790258923D3CA4E1D7E11F604B32D0EFE18B449B
057CC736ECE0BF820A42555426DB88348500000006A26863726564426C6F6244313231326B686D61632D736563726
574585051144A81FFFCED51BB81932851ABB84D17325C764C95C5016B5ED0F80FA763728F7D410B39103FC502E115
D7DAA326D6D0FA59A06BB20DFB4DA9AE31F1E3840D520FF29A01CD4B98D7B555730C2896730358473045022033626
FC0675341C2EBC5D5A049D739F517110F749685E12C3DABCB2EB129E03022100D3CE29E77F8EBB3A0B63D858E56A
CE6B3A8702BF5113BBE7A081D352F844914F04A1626964506D696B65406578616D706C652E636F6D'
}
```

Inner Authenticator response details following the first byte indicating success(0x00):

```
A401A262696458301C1AEF38C820EE448C7BF0EE8512BF9A813A96B73019E028E14270EB5AD74D2383EE72AC40DDA
BF53D3222CEBEA861D464747970656A7075626C69632D6B65790258923D3CA4E1D7E11F604B32D0EFE18B449B057C
C736ECE0BF820A42555426DB88348500000006A26863726564426C6F6244313231326B686D61632D7365637265745
85051144A81FFFCED51BB81932851ABB84D17325C764C95C5016B5ED0F80FA763728F7D410B39103FC502E115D7DA
A326D6D0FA59A06BB20DFB4DA9AE31F1E3840D520FF29A01CD4B98D7B555730C2896730358473045022033626FC06
75341C2EBC5D5A049D739F517110F749685E12C3DABCB2EB129E03022100D3CE29E77F8EBB3A0B63D858E56ACE6B
3A8702BF5113BBE7A081D352F844914F04A1626964506D696B65406578616D706C652E636F6D
```

Textual representation of the preceding map:

```
{
  1: {
    id:
h'1C1AEF38C820EE448C7BF0EE8512BF9A813A96B73019E028E14270EB5AD74D2383EE72AC40DDABF53D3222CEBEA
861D4',
    type: "public-key"
  },
  2:
h'3D3CA4E1D7E11F604B32D0EFE18B449B057CC736ECE0BF820A42555426DB88348500000006A26863726564426C6
F6244313231326B686D61632D736563726574585051144A81FFFCED51BB81932851ABB84D17325C764C95C5016B5E
D0F80FA763728F7D410B39103FC502E115D7DAA326D6D0FA59A06BB20DFB4DA9AE31F1E3840D520FF29A01CD4B98D
7B555730C289673',
  3:
h'3045022033626FC0675341C2EBC5D5A049D739F517110F749685E12C3DABCB2EB129E03022100D3CE29E77F8EB
B3A0B63D858E56ACE6B3A8702BF5113BBE7A081D352F844914F',
  4: {
    id: h'6D696B65406578616D706C652E636F6D'
  }
}
```

4.5 CTAPCBOR_RPC_COMMAND_GET_CREDENTIALS

4.5.1 Request

Full Request (a CBOR map):

```
0xAA67636F6D6D616E640965666C616773006774696D656F7574006D7472616E73616374696F6E496450D68633AC1
123364E9709E201FD05693C7566696C7465724879627269645472616E73706F7274F4647270496468637461702E64
6576782061757468656E74696361746F72496E666F4C6F676F526571756573745479706500781F706C7567696E417
5746F66696C6C5363656E6172696F537570706F72746564F46F75765472616E73616374696F6E4964500000000000
000000000000000000071746869726450617274795061796D656E74F4
```

Textual representation of the CBOR encoding:

```
{
  command: 9,
```



```
flags: 0,
timeout: 0,
transactionId: h'D68633AC1123364E9709E201FD05693C',
filterHybridTransport: false,
rpId: "ctap.dev",
authenticatorInfoLogoRequestType: 0,
pluginAutofillScenarioSupported: false,
uvTransactionId: h'00000000000000000000000000000000',
thirdPartyPayment: false
}
```

4.5.2 Response

Full Response (a CBOR map):

```
0x000081AA00040158201C337E7A9F27200C98BE1E0337A501C02CBFA7215A85EA8CD9E187A1E97EFDDDD02A26
2696468637461702E646576646E616D6574576562417574686E20546573742053657276657203A36269644F62
6F62406578616D706C652E636F6D646E616D656F626F62406578616D706C652E636F6D6B646973706C61794E6
16D6569426F6220536D69746804F505F4066D57696E646F77732048656C6C6F075908D23C7376672076696577
426F783D22302030203339203339222066696C6C3D226E6F6E652220786D6C6E733D22687474703A2F2F77777
72E77332E6F72672F323030302F737667223E3C726563742077696474683D22343922206865696768743D2234
39222072783D2232222066696C6C3D227768697465222F3E3C6D61736B2069643D226D61736B305F313137383
15F31363230313422207374796C653D226D61736B2D747970653A6C756D696E616E636522206D61736B556E69
74733D227573657253706163654F6E5573652220783D22382220793D2238222077696474683D2233332220686
5696768743D2233323323E3C7061746820643D224D34312038483856343148343156385A222066696C6C3D2277
68697465222F3E3C2F6D61736B3E3C67206D61736B3D2275726C28236D61736B305F31313738315F313632303
1342922207472616E73666F726D3D227472616E736C617465282D352E352C2D3529223E3C706174682066696C
6C2D72756C653D226576656E6F64642220636C69702D72756C653D226576656E6F64642220643D224D31372E3
632352031362E32354331372E3632352031322E383332372032302E333935322031302E303632352032332E38
3132352031302E303632354332362E393137342031302E303632352032392E343838332031322E33343935203
2392E393332312031352E333330384333302E3533352031342E393734342033312E313837322031342E363932
352033312E383736312031342E343937384333312E303732372031302E373833312032372E373637362038203
2332E3831323520384331392E3235363120382031352E353632352031312E363933362031352E353632352031
362E32354331352E353632352032302E383036342031392E323536312032342E352032332E383132352032342
E354332342E353932312032342E352032352E333436352032342E333931392032362E303631352032342E3138
39374332352E393339332032332E3632352032352E3837352032332E303338382032352E3837352032322E343
337354332352E3837352032322E333138372032352E383737352032322E323030342032352E38383235203232
2E303832384332352E323335332032322E333132352032342E353338352032322E343337352032332E3831323
52032322E343337354332302E333935322032322E343337352031372E3632352031392E363637332031372E36
32352031362E32355A4D31332E352032362E353632354832362E393738374332372E343237382032372E33333
8382032372E393939382032382E303335322032382E3636382032382E3632354831332E354331322E39353320
32382E3632352031322E343238342032382E383432342031322E303431362032392E323239314331312E36353
4382032392E363135382031312E343337352033302E313430352031312E343337352033302E36383735433131
2E343337352033322E3934382031322E333333372033342E393538332031342E323536322033362E343331384
331362E32303822033372E393237392031392E333130342033382E393337352032332E383132352033382E39
3337354332362E323832372033382E393337352032382E333331342033382E363333352033302033382E31313
2315634302E323630324332382E323230392034302E373432362032362E313538362034312032332E38313235
2034314331392E303333342034312031352E343332342033392E393331382031332E303031362033382E30363
8384331302E353431332033362E3138333220392E333735203332E3535323920392E3337352033302E363837
3543392E3337352032392E3539333520392E38303935392032382E353434332031302E353833322032372E373
730374331312E333536382032362E393937312031322E3430362032362E353632352031332E352032362E3536
32355A4D32382E323839362032342E34394332382E303631372032332E383435372032372E393337352032332
E31352332032372E393337352032322E34334332372E393337352031392E303136392033302E373038332031
362E32352033342E313235362031362E32354333372E353432382031362E32352034302E333133312031392E3
03136392034302E333133312032322E34334334302E333133312032352E3138382033382E353034312032372E
353234312033362E303036362032382E333139324C33372E393433372033302E303034324333382E333438342
033302E333536312033382E333438342033302E393833392033372E393433372033312E333335374C33352E31
353639203332E37364C33372E393335372033362E32334433382E333132332033362E353639332033382E3
33331392033372E313531322033372E393738382033372E353131314C33342E383134312034302E3733353243
33342E343436332034312E31303939203332E383334342034312E30383333203332E353030352034302E363
738324C33322E323634362033392E31373836433322E313334332033392E303230362033322E303633312033
382E383232322033322E303633312033382E363137344C33322E303632352033322E333531335632382E32353
834433322E303034332032382E323337382033312E393436362032382E323136322033312E38383933203238
2E3139344333302E323133332032372E353434382032382E383930342032362E313837372032382E323839362
032342E34395A4D33342E313235362032322E34334333352E323634372032322E34332033362E313838312032
```

312E353037372033362E313838312032302E33374333362E313838312031392E323332332033352E323634372
031382E33312033342E313235362031382E33314333322E393836352031382E33312033322E30363331203139
2E323332332033322E303633312032302E33374333322E303633312032312E353037372033322E39383635203
2322E34332033342E313235362032322E343335A222066696C6C3D22626C61636B222066696C6C2D6F70616369
74793D22302E38393536222F3E3C2F673E3C2F7376673E0008F40910

Textual representation of the CBOR encoding after the HRESULT:

```
[
  {
    0: 4,
    1: h'1C337E7A9F27200C98BE1E0337A501C02CBFA7215A85EA8CD9E187A1E97EFDDD',
    2: {
      id: "ctap.dev",
      name: "WebAuthn Test Server"
    },
    3: {
      id: h'626F62406578616D706C652E636F6D',
      name: "bob@example.com",
      displayName: "Bob Smith"
    },
    4: true,
    5: false,
    6: "Windows Hello",
    7:
h'3C7376672076696577426F783D22302030203339203339222066696C6C3D226E6F6E652220786D6C6E733D2
2687474703A2F7777772E77332E6F72672F323030302F737667223E3C726563742077696474683D22343922
206865696768743D223439222072783D223222066696C6C3D227768697465222F3E3C6D61736B2069643D226
D61736B305F31313738315F31363230313422207374796C653D226D61736B2D747970653A6C756D696E616E63
6522206D61736B556E6974733D227573657253706163654F6E5573652220783D22382220793D2238222077696
474683D223332206865696768743D223333223E3C7061746820643D224D3431203848385634314834315638
5A222066696C6C3D227768697465222F3E3C2F6D61736B3E3C67206D61736B3D2275726C28236D61736B305F3
1313738315F3136323031342922207472616E73666F726D3D227472616E736C617465282D352E352C2D352922
3E3C706174682066696C6C2D72756C653D226576656E6F64642220636C69702D72756C653D226576656E6F646
42220643D224D31372E3632352031362E32354331372E3632352031322E383332372032302E33393532203130
2E303632352032332E383132352031302E303632354332362E393137342031302E303632352032392E3438383
32031322E333439352032392E393332312031352E333330384333302E3533352031342E393734342033312E31
3837322031342E363932352033312E383736312031342E343937384333312E303732372031302E37383331203
2372E3736373620382032332E3831323520384331392E3235363120382031352E353632352031312E36393336
2031352E353632352031362E32354331352E353632352032302E383036342031392E323536312032342E35203
2332E383132352032342E354332342E353932312032342E352032352E333436352032342E333931392032362E
303631352032342E313839374332352E393339332032332E3632352032352E3837352032332E3033383820323
52E3837352032322E343337354332352E3837352032322E333138372032352E383737352032322E3230303420
32352E383832352032322E303832384332352E32333532032322E333132352032342E353338352032322E343
337352032332E383132352032322E343337354332302E333935322032322E343337352031372E363235203139
2E363637332031372E3632352031362E32355A4D31332E352032362E353632354832362E393738374332372E3
43237382032372E333338382032372E393939382032382E303335322032382E3636382032382E363235483133
2E354331322E3935332032382E3632352031322E343238342032382E383432342031322E303431362032392E3
23239314331312E363534382032392E363135382031312E343337352033302E313430352031312E3433373520
33302E363837354331312E343337352033322E3934382031322E333333372033342E393538332031342E32353
6322033362E343331384331362E323038322033372E393237392031392E333130342033382E39333735203233
2E383132352033382E393337354332362E323832372033382E393337352032382E333331342033382E3633333
5203302033382E313132315634302E323630324332382E323230392034302E373432362032362E3135383620
34312032332E383132352034314331392E303333342034312031352E343332342033392E393331382031332E3
03031362033382E303638384331302E353431332033362E3138333220392E3337352033332E3535323920392E
3337352033302E3638373543392E3337352032392E3539333520392E38303935392032382E353434332031302
E353833322032372E373730374331312E333536382032362E393937312031322E3430362032362E3536323520
31332E352032362E353632355A4D32382E323839362032342E34394332382E303631372032332E38343537203
2372E393337352032332E313532332032372E393337352032322E34334332372E393337352031392E30313639
2033302E373038332031362E32352033342E313235362031362E32354333372E353432382031362E323520343
02E333133312031392E303136392034302E3331333120323232E3433434302E333133312032352E33138382033
382E353034312032372E353234312033362E303036362032382E333139324C33372E393433372033302E30303
4324333382E333438342033302E333536312033382E333438342033302E393833392033372E39343337203331
2E333335374C33352E313536392033332E37364C33372E393335372033362E3233344333382E3331323320333
62E353639332033382E333331392033372E313531322033372E393738382033372E353131314C33342E383134
312034302E373335324333342E343436332034312E313039392033332E383334342034312E303833332033332
E353030352034302E363738324C33322E323634362033392E313738364333322E313334332033392E30323036
2033322E303633312033382E383232322033322E303633312033382E363137344C33322E303632352033322E3
```

```
33531335632382E32353834433322E303034332032382E323337382033312E393436362032382E3231363220
33312E383839332032382E3139344333302E323133332032372E353434382032382E383930342032362E31383
7372032382E323839362032342E34395A4D33342E313235362032322E34334333352E323634372032322E3433
2033362E313838312032312E353037372033362E313838312032302E33374333362E313838312031392E32333
2332033352E323634372031382E33312033342E313235362031382E33314333322E393836352031382E333120
33322E303633312031392E323332332033322E303633312032302E33374333322E303633312032312E3530373
72033322E393836352032322E34332033342E313235362032322E34335A222066696C6C3D22626C61636B2220
66696C6C2D6F7061636974793D22302E38393536222F3E3C2F673E3C2F7376673E00',
  8: false,
  9: 16
}
]
```

4.6 CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST

4.6.1 Request

Full Request (a CBOR map):

```
A667636F6D6D616E640C65666C616773006774696D656F7574006D7472616E73616374696F6E4964500000000
0000000000000000000000000782061757468656E74696361746F72496E666F4C6F676F526571756573745479
70650171746869726450617274795061796D656E74F4
```

Textual representation of the CBOR encoding:

```
{
  command: 12,
  flags: 0,
  timeout: 0,
  transactionId: h'00000000000000000000000000000000',
  authenticatorInfoLogoRequestType: 1,
  thirdPartyPayment: false
}
```

4.6.2 Response

Full Response (a CBOR map):

```
0x000081A50101025008987058CADC4B81B6E130DE50DCBE96036D57696E646F77732048656C6C6F045901AB3
C7376672069643D224C617965725F312220786D6C6E733D22687474703A2F2F7777772E77332E6F72672F3230
30302F737667222076696577426F783D223020302032353620323536223E3C646566733E3C7374796C653E2E6
36C732D317B66696C6C3A233030373864343B7374726F6B652D77696474683A3070783B7D3C2F7374796C653E
3C2F646566733E3C7265637420636C6173733D22636C732D312220783D2232342E32352220793D2232342E323
5222077696474683D2239382E333522206865696768743D2239382E3335222F3E3C7265637420636C6173733D
22636C732D312220783D223133332E342220793D2232342E3235222077696474683D2239382E3335222068656
96768743D2239382E3335222F3E3C7265637420636C6173733D22636C732D312220783D2232342E3235222079
3D223133332E34222077696474683D2239382E333522206865696768743D2239382E3335222F3E3C726563742
0636C6173733D22636C732D312220783D223133332E342220793D223133332E34222077696474683D2239382E
333522206865696768743D2239382E3335222F3E3C2F7376673E05F4
```

Textual representation of the CBOR encoding after the HRESULT:

```
[
  {
    1: 1,
  }
]
```

```
2: h'08987058CADC4B81B6E130DE50DCBE96',
3: "Windows Hello",
4:
h'3C7376672069643D224C617965725F312220786D6C6E733D22687474703A2F2F7777772E77332E6F72672F3
23030302F737667222076696577426F783D223020302032353620323536223E3C646566733E3C7374796C653E
2E636C732D317B66696C6C3A233030373864343B7374726F6B652D77696474683A3070783B7D3C2F7374796C6
53E3C2F646566733E3C7265637420636C6173733D22636C732D312220783D2232342E32352220793D2232342E
3235222077696474683D2239382E333522206865696768743D2239382E3335222F3E3C7265637420636C61737
33D22636C732D312220783D223133332E342220793D2232342E3235222077696474683D2239382E3335222068
65696768743D2239382E3335222F3E3C7265637420636C6173733D22636C732D312220783D2232342E3235222
0793D223133332E34222077696474683D2239382E333522206865696768743D2239382E3335222F3E3C726563
7420636C6173733D22636C732D312220783D223133332E342220793D223133332E34222077696474683D22393
82E333522206865696768743D2239382E3335222F3E3C2F7376673E',
5: false
}
]
```

5 Security

5.1 Security Considerations for Implementers

For information, see [\[W3C-WebAuthPKC3\]](#), section 13.

5.2 Index of Security Parameters

None.

6 Appendix A: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

The terms "earlier" and "later", when used with a product version, refer to either all preceding versions or all subsequent versions, respectively. The term "through" refers to the inclusive range of versions. Applicable Microsoft products are listed chronologically in this section.

Windows Client

- Windows 10 v1809 operating system
- Windows 11 operating system

Windows Server

- Windows Server v1809 operating system
- Windows Server 2019 operating system
- Windows Server 2022 operating system
- Windows Server 2025 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

[<1> Section 2.2.1](#): This is the Microsoft Windows API version on Windows systems. See WebAuthNGetApiVersionNumber in [\[MSFT-WebAuthnAPIS\]](#).

[<2> Section 2.2.1](#): Windows 11, version 24H2 operating system and later and Windows Server 2025 and later with [\[MSKB-5065789\]](#) support CTAPCBOR_RPC_COMMAND_GET_CREDENTIALS.

[<3> Section 2.2.1](#): Windows 11, version 24H2 and later and Windows Server 2025 and later with [\[MSKB-5065789\]](#) support CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST.

[<4> Section 2.2.1.1](#): For example, Windows Hello on Windows systems.

7 Change Tracking

This section identifies changes that were made to this document since the last release. Changes are classified as Major, Minor, or None.

The revision class **Major** means that the technical content in the document was significantly revised. Major changes affect protocol interoperability or implementation. Examples of major changes are:

- A document revision that incorporates changes to interoperability requirements.
- A document revision that captures changes to protocol functionality.

The revision class **Minor** means that the meaning of the technical content was clarified. Minor changes do not affect protocol interoperability or implementation. Examples of minor changes are updates to clarify ambiguity at the sentence, paragraph, or table level.

The revision class **None** means that no new technical changes were introduced. Minor editorial and formatting changes may have been made, but the relevant technical content is identical to the last released version.

The changes made to this document are listed in the following table. For more information, please contact dochelp@microsoft.com.

Section	Description	Revision class
2.2.1 WebAuthN_Channel Request Message	Added the CTAPCBOR_RPC_COMMAND_GET_CREDENTIAL and CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST RPC commands.	Major
2.2.1 WebAuthN_Channel Request Message	Added the CTAPCBOR_RPC_COMMAND_GET_CREDENTIAL and CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST RPC commands.	Major
2.2.1 WebAuthN_Channel Request Message	Added the CTAPCBOR_RPC_COMMAND_GET_CREDENTIAL and CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST RPC commands.	Major
2.2.1 WebAuthN_Channel Request Message	Added the CTAPCBOR_RPC_COMMAND_GET_CREDENTIAL and CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST RPC commands.	Major
2.2.1 WebAuthN_Channel Request Message	Added the CTAPCBOR_RPC_COMMAND_GET_CREDENTIAL and CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST RPC commands.	Major
2.2.1 WebAuthN_Channel Request Message	Added new fields to WebAuthN_Channel Request Message.	Major
2.2.1.1 webAuthNPara Map	Added new fields to webAuthNPara Map.	Major
2.2.1.2 CTAPCBOR_CMD_MAKE_CREDENTIAL	Updated request example.	Major

Section	Description	Revision class
Request		
2.2.1.3 CTAPCBOR_CMD_GET_ASSERTION Request	Updated request example.	Major
2.2.2 WebAuthN_Channel Response Message	Added the CTAPCBOR_RPC_COMMAND_GET_CREDENTIALS and CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST RPC commands.	Major
2.2.2.1 CTAPCBOR_RPC_COMMAND_WEB_AUTHN Response Map	Added new fields to CTAPCBOR_RPC_COMMAND_WEB_AUTHN Response Map.	Major
2.2.2.1.1 CTAP MakeCredential Response	Updated response example.	Major
2.2.2.1.2 CTAP GetAssertion Response	Updated response example.	Major
2.2.2.2 CTAPCBOR_RPC_COMMAND_GET_CREDENTIALS Response Map	Added description, fields, and example response.	Major
2.2.2.3 CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST Response Map	Added description, fields, and example response.	Major
4.4.1.1 Request	Updated request sample.	Major
4.4.1.2 Response	Updated response sample.	Major
4.4.2.1 Request	Updated request sample.	Major
4.4.2.2 Response	Updated response sample.	Major
4.5 CTAPCBOR_RPC_COMMAND_GET_CREDENTIALS	Added a new section.	Major
4.5.1 Request	Added request sample.	Major
4.5.2 Response	Added response sample.	Major
4.6 CTAPCBOR_RPC_COMMAND_GET_AUTHENTICATOR_LIST	Added a new section.	Major
4.6.1 Request	Added request sample.	Major
4.6.2 Response	Added response sample.	Major

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